Lecture#3

ER-Diagram

Lecture2 Review Questions

1-Match the following terms and definitions:

---------- Information Engineering

----------- Entities

-----------Attribute

------------ Relationships

a) property or characteristic of an entity or relationship type

b) instance–link between entities.

c) A data-oriented methodology to create and maintain information systems.

d) instance–person, place, object, event, concept

Attribute

Property or characteristic of an entity

A Good Attribute Name is:

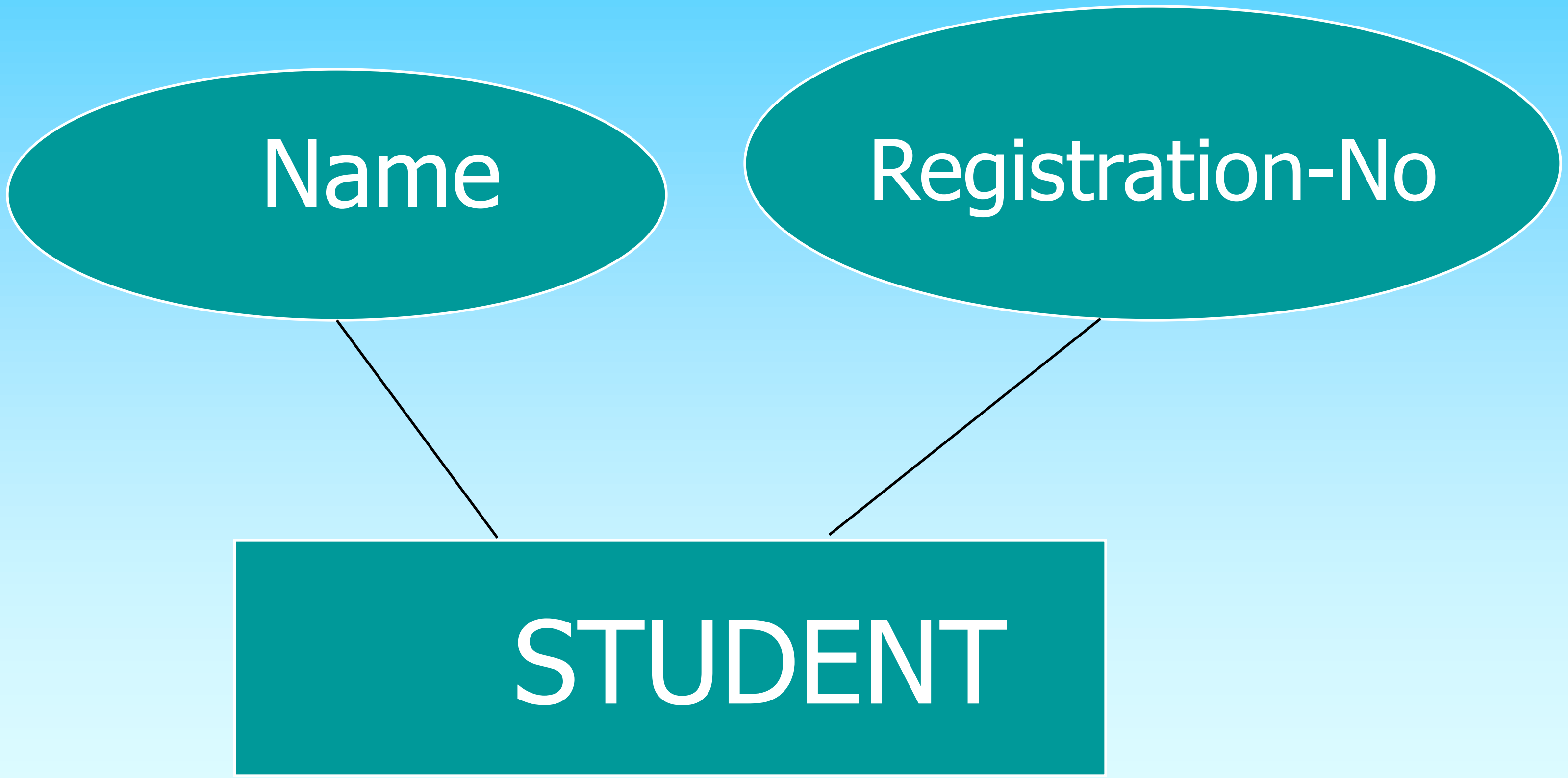
Related to business characteristics

Meaningful and self-documenting

Unique

Readable

Attribute



Classifications of attributes

- Required versus Optional Attributes
- Simple versus Composite Attribute
- Single-Valued versus Multi-valued Attribute
- Stored versus Derived Attributes
- Identifier Attributes

Required attribute

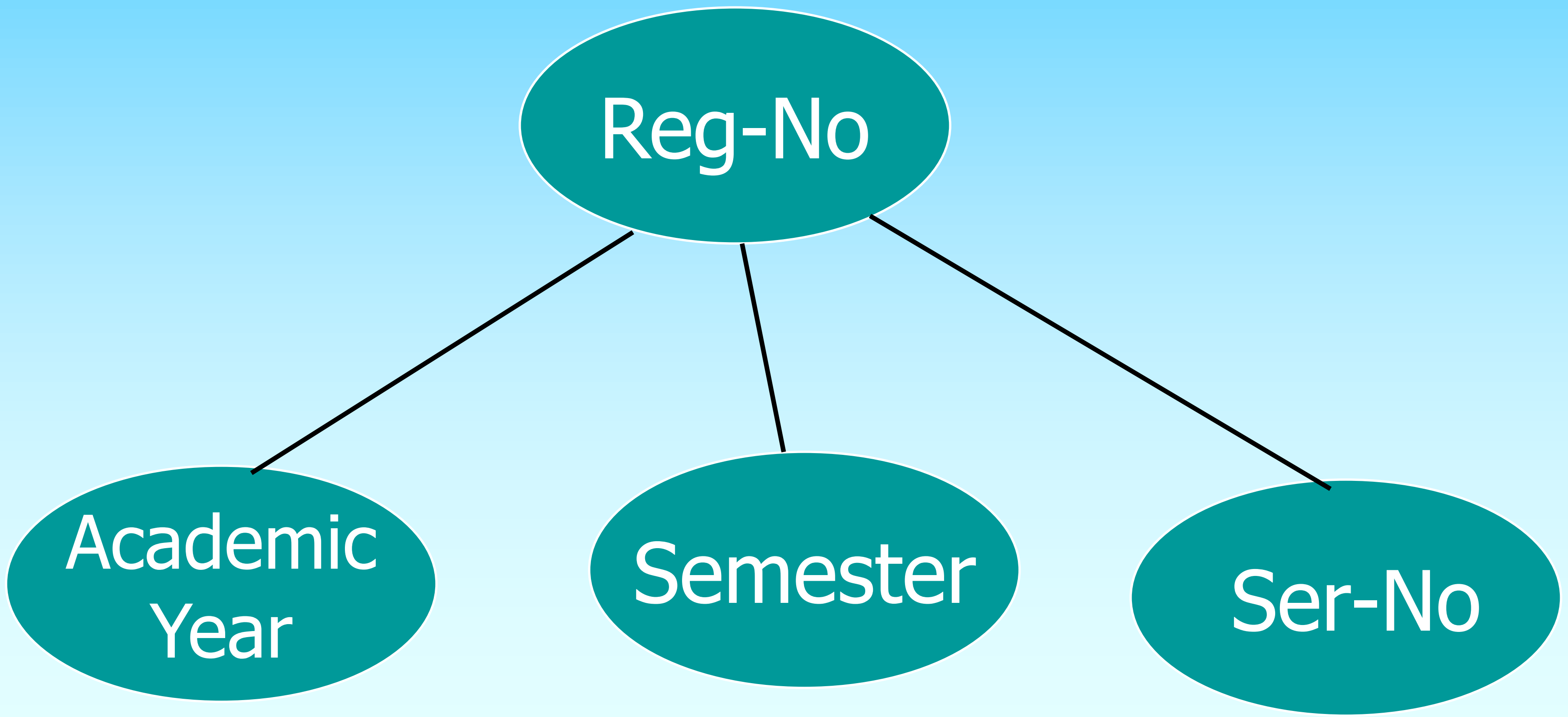
An attribute that **must have a value** for every entity instance.

Optional attribute

An attribute that may not have a value for every entity instance

A composite attribute

An attribute broken into component parts



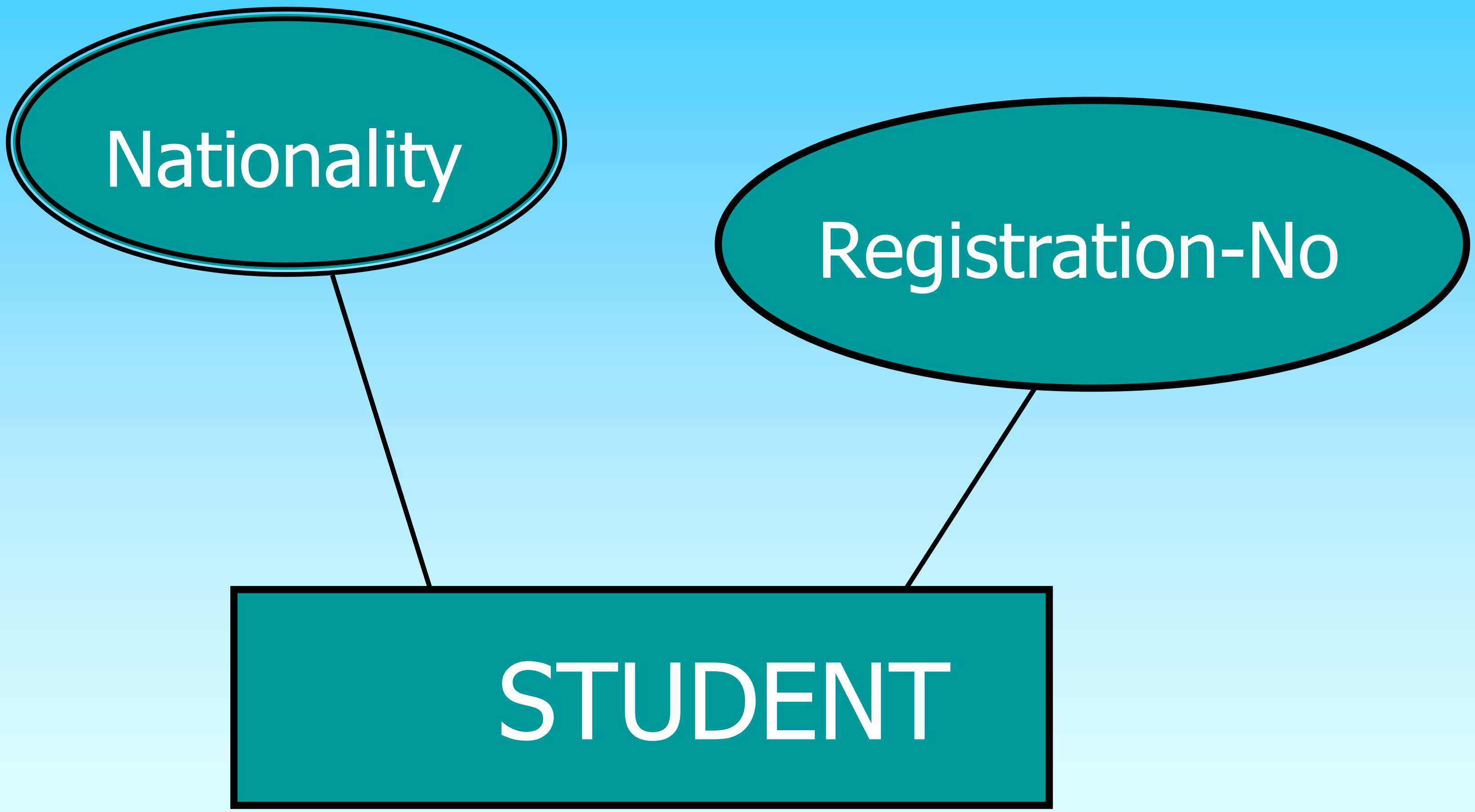
Single-valued Attribute

An attribute that can have only one value

Multi-valued Attribute

An attribute that can have more than one value

Multi-Valued Attribute



Derived attribute

* Should have a value (not optional).

* Should be single valued.

* Should be of changeable value.

Identifier attribute

Identifier is an attribute that uniquely identify the instance of the entity type

Ex.

Student_Id

National number

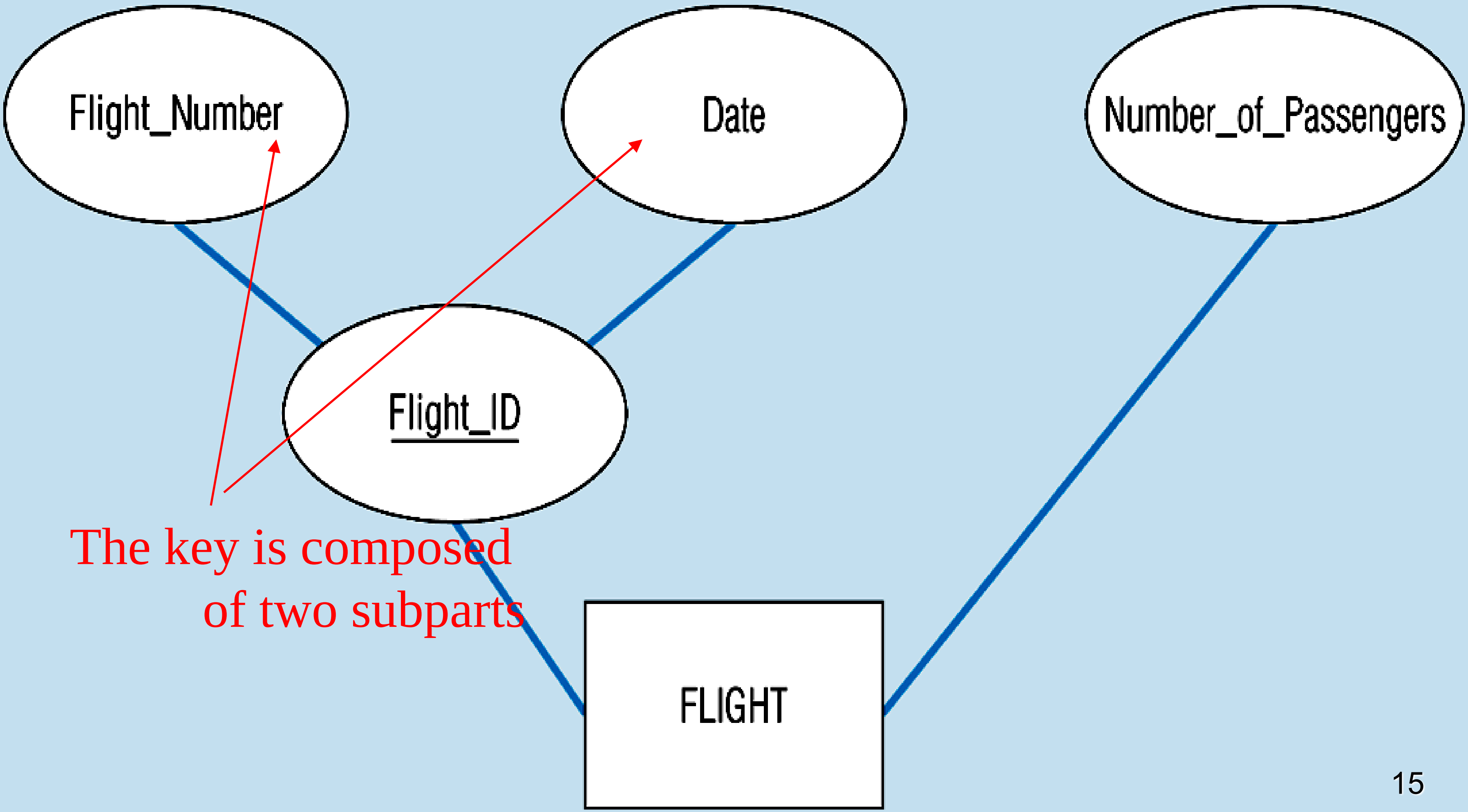
Characteristics of Identifiers

Will not change in value - Address for example

Will not be null – must have a value

Substitute new, simple keys for long, composite keys

Composite key attribute



The key is composed of two subparts

Computer training center ERD

Future is a computer training center, Develop an ER diagram for the following:

A database used to contain information about instructors, trainee, and course. The following information is to be included:

For each Instructor:

- instructor id (unique).
- instructor name (first, middle, last).
- birthday.
- age.
- major.
- Qualifications (several per instructor).
- address (city, street).
- telephone num (several per insteructor).

Continued

For each Trainee:

trainee id (unique).

trainee name (first, middle, last).

birthday.

age.

major.

address (city, street).

telephone num (several per trainee).

For each course:

course id.

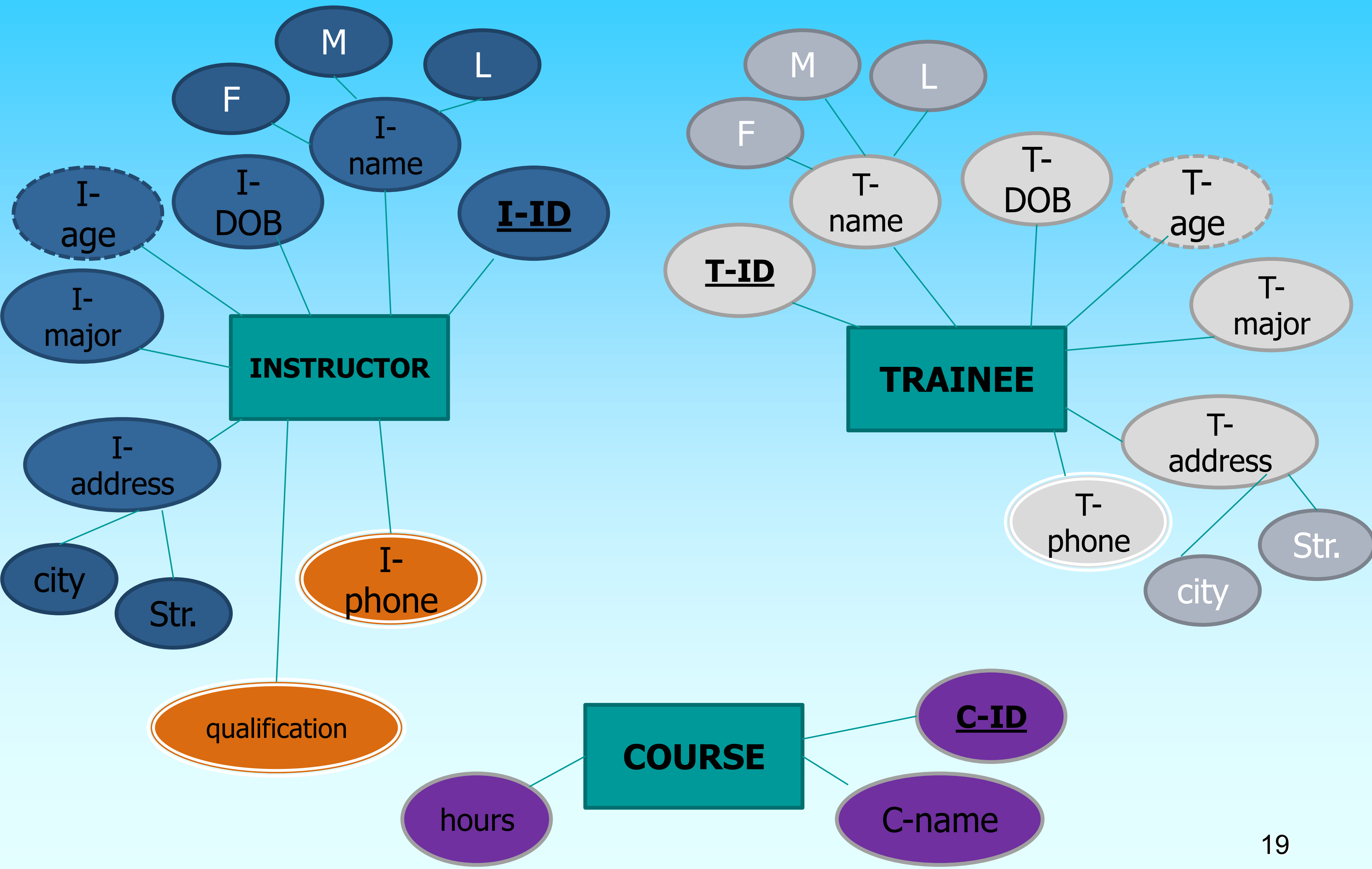
course name.

hours.

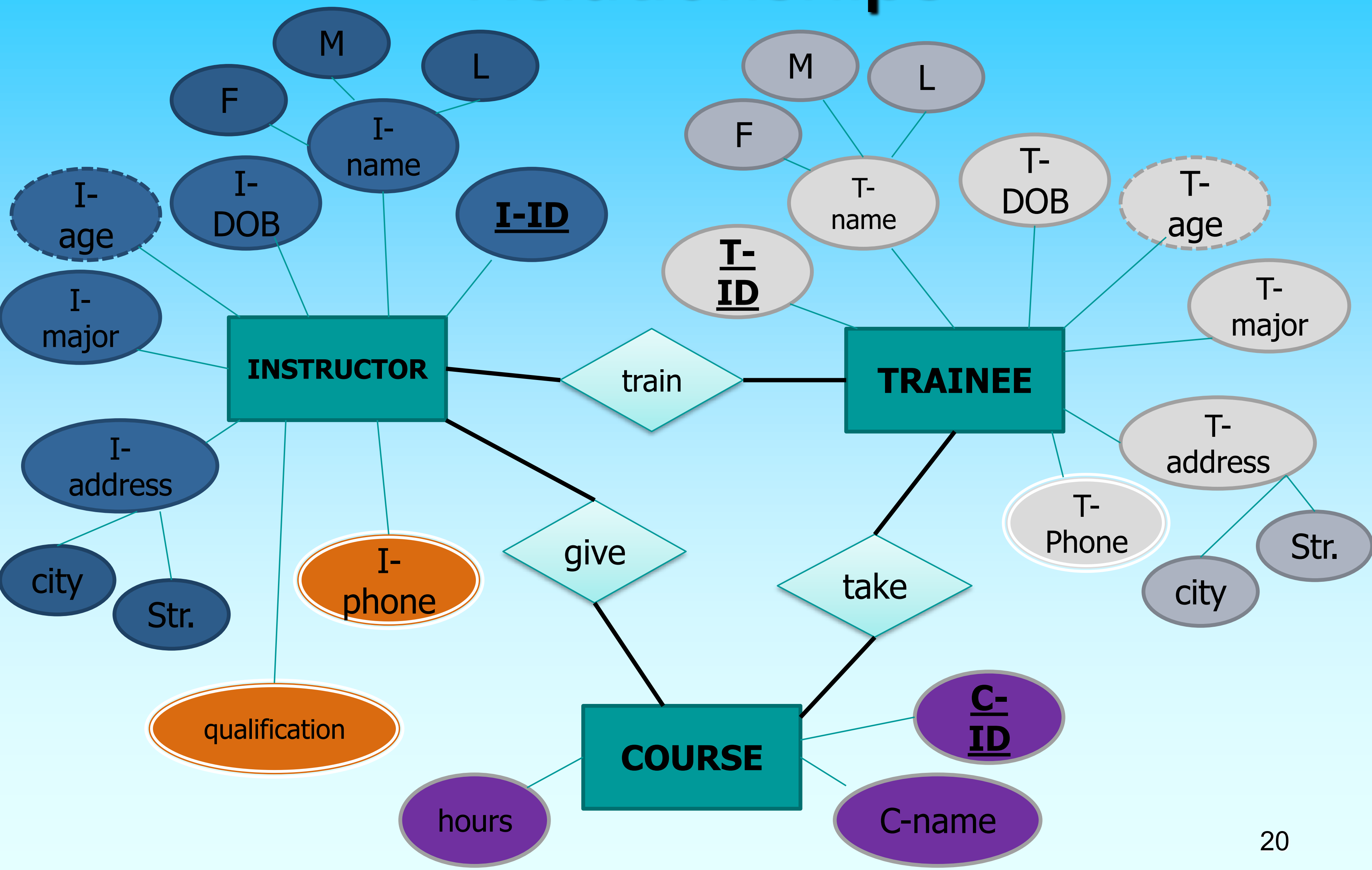
Continued

- Instructor give more than one course.
- Each course is given by one or more instructor.
- Instructor train many trainees.
- Each trainee must be trained by an instructor.
- Each Trainee should enroll at least in one course.

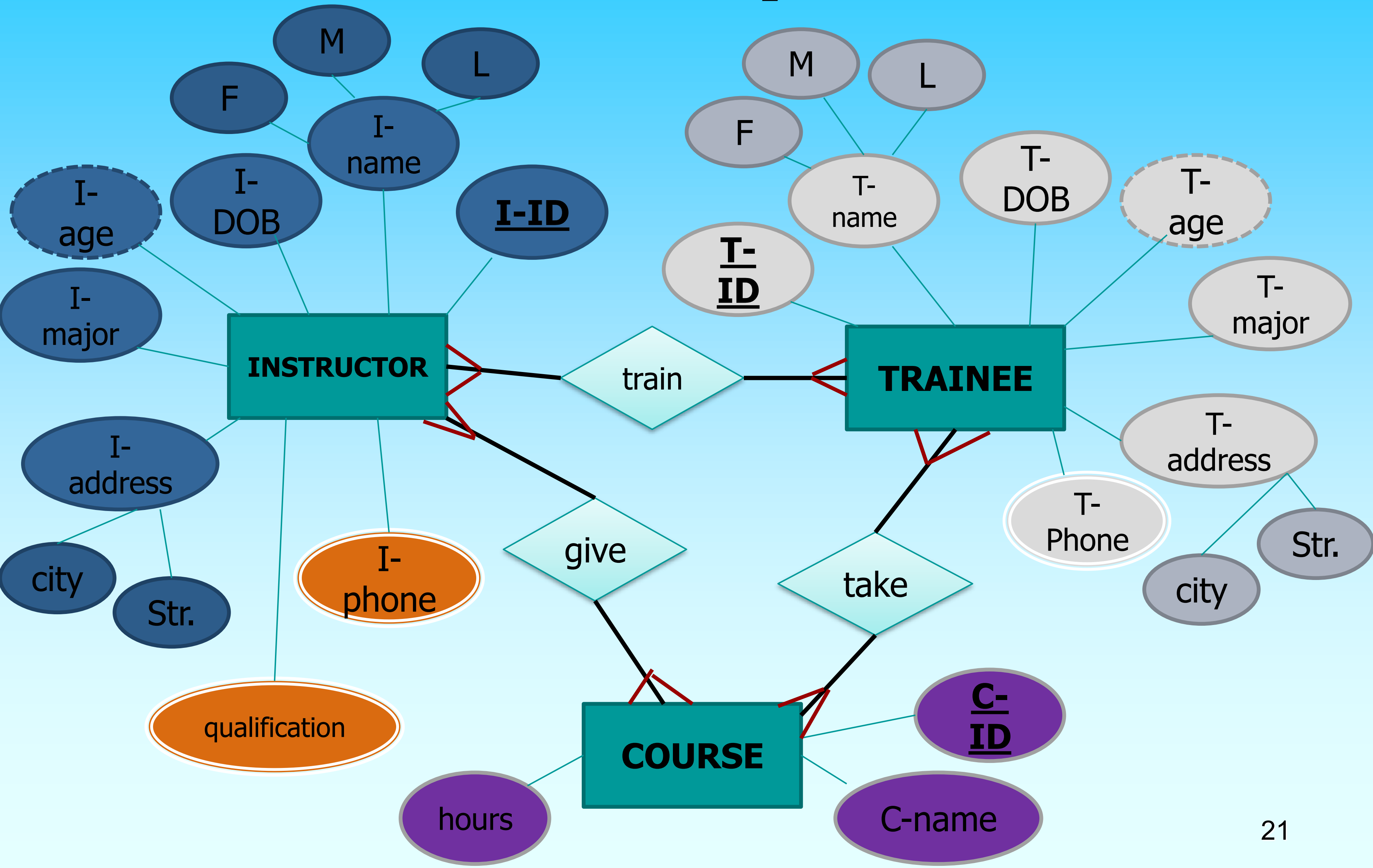
Entities and Attributes



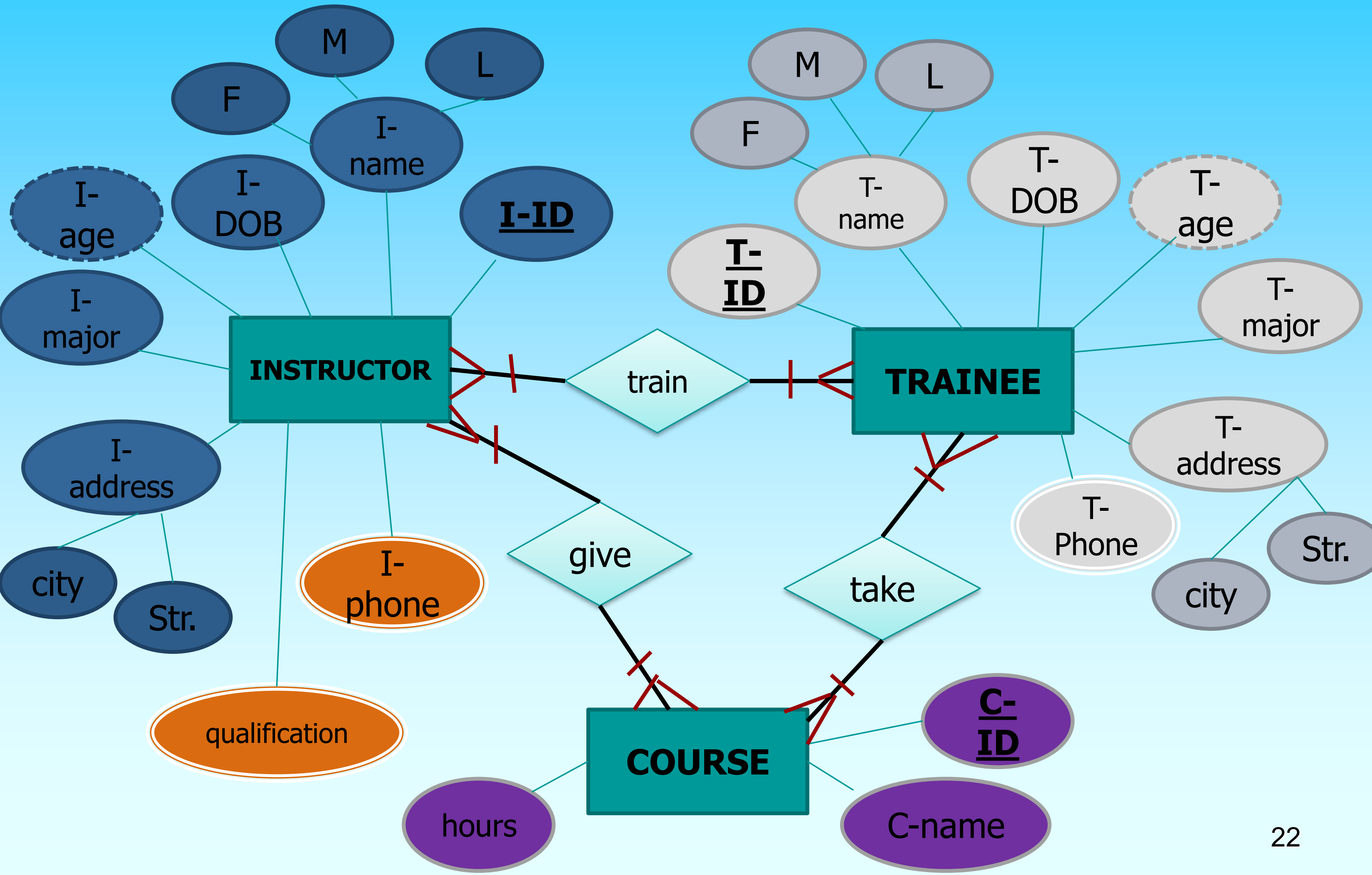
Relationships



Cardinality ratios



Relationship Cardinality



problem

A company wants to create a database to store information about: Employee, Department and Project. The following information is to be included:

Employee: E-id, name, address.

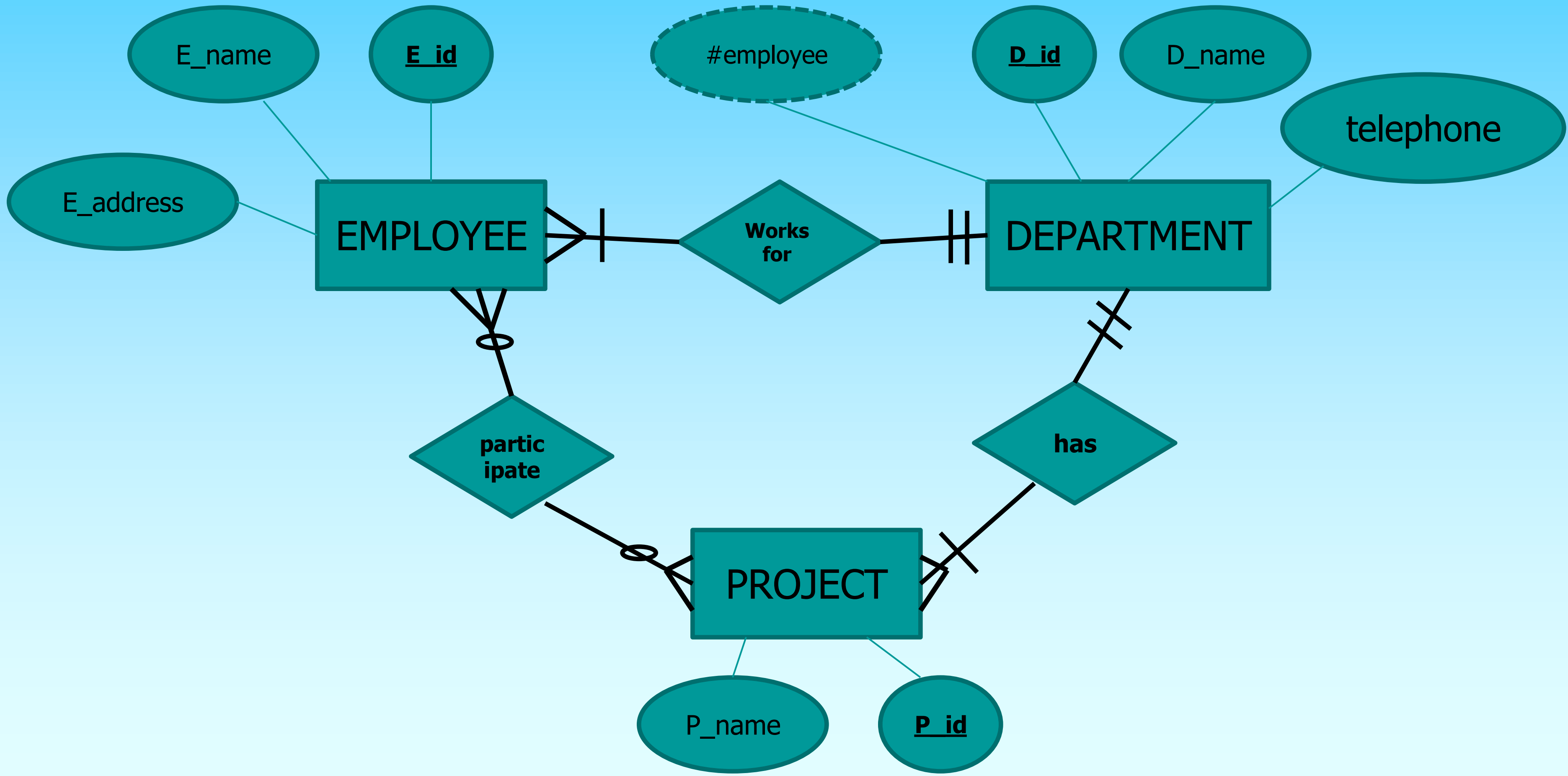
Department: id, name, telephone, number of employee.

Project: id, name.

Draw an ERD to present the following:

- Employee must works for one department.
- Employee may participates in more than one project.
- Each department should have at least project.
- Each Project should belong to one department.

Answer


E_name
E_id
#employee
D_id
D_name
telephone
E_address
EMPLOYEE
Works
for
DEPARTMENT
partic
ipate
has
PROJECT
P_name
P_id

example

- **Online Bookstore Database:**
- Business Rules:
  - A Book can be written by one or more Authors.
  - A Customer can place multiple Orders.
  - An Order can contain one or more Books, and each Book can be in multiple Orders.
  - Each Order is associated with only one Payment.
  - Each Order is shipped to only one Customer.

example

- **University Registration System:**
- Business Rules:
  - A Student can enroll in multiple Courses, and each Course can have multiple Students.
  - Each Course is taught by exactly one Instructor.
  - Each Course belongs to one Department, but a Department can have multiple Courses.
  - Each Student's enrollment is unique for each Course; a Student cannot enroll in the same Course multiple times.

Draw an ERD for each of the following situations. (If you believe that you need to make additional assumptions, clearly state them for each situation.) Draw the same situation using the tool you have been told to use in the course.

a. A company has a number of employees. The attributes of EMPLOYEE include Employee_ID (identifier), Name, Address, and Birthdate. The company also has several projects. Attributes of PROJECT include Project_ID (identifier), Project_Name, and Start_Date. Each employee may be assigned to one or more projects, or may not be assigned to a project. A project must have at least one employee assigned and may have any number of employees assigned. An employee's billing rate may vary by project, and the company wishes to record the applicable billing rate (Billing_Rate) for each employee when assigned to a particular project. Do the attribute names in this description follow the guidelines for naming attributes? If not, suggest better names.

b. A university has a large number of courses in its catalog. Attributes of COURSE include Course_Number (identifier), Course_Name, and Units. Each course may have one or more different courses as prerequisites or may have no prerequisites. Similarly, a particular course may be a prerequisite for any number of courses or may not be prerequisite for any other course. Provide a good definition of COURSE. Why is your definition a good one?

c. A laboratory has several chemists who work on one or more projects. Chemists also may use certain kinds of equipment on each project. Attributes of CHEMIST include Employee_ID (identifier), Name, and Phone_No. Attributes of PROJECT include Project_ID (identifier) and Start_Date. Attributes of EQUIPMENT include Serial_No and Cost. The organization wishes to record Assign_Date—that is, the date when a given equipment item was assigned to a particular chemist working on a specified project. A chemist must be assigned to at least one project and one equipment item. A given equipment item need not be assigned, and a given project need not be assigned either a chemist or an equipment item. Provide good definitions for all of the relationships in this situation.

d. A college course may have one or more scheduled sections, or may not have a scheduled section. Attributes of COURSE include Course_ID, Course_Name, and Units. Attributes of SECTION include Section_Number and Semester_ID. Semester_ID is composed of two parts: Semester and Year. Section_Number is an integer (such as "1" or "2") that distinguishes one section from another for the same course but does not uniquely identify a section. How did you model SECTION? Why did you choose this way versus alternative ways to model SECTION?

e. A hospital has a large number of registered physicians. Attributes of PHYSICIAN include Physician_ID (the identifier) and Specialty. Patients are admitted to the hospital by physicians. Attributes of PATIENT include Patient_ID (the identifier) and Patient_Name. Any patient who is admitted must have exactly one admitting physician. A physician may optionally admit any number of patients. Once admitted, a given patient must be treated by at least one physician. A particular physician may treat any number of patients, or may not treat any patients. Whenever a patient is treated by a physician, the hospital wishes to record the details of the treatment (Treatment_Detail). Components of Treatment_Detail include Date, Time, and Results. Did you draw more than one relationship between physician and patient? Why or why not?

Assignment **: CREATING AN ENTITY-RELATIONSHP DIAGRAM.**

UPS prides itself on having up-to-date information on the processing and current location of each shipped item. To do this, UPS relies on a company-wide information system. Shipped items are the heart of the UPS product tracking information system. Shipped items can be characterized by item number (unique), weight, dimensions, insurance amount, destination, and final delivery date.

Shipped items are received into the UPS system at a single retail center. Retail centers are characterized by their type, unique ID, and address. Shipped items make their way to their destination via one or more standard UPS transportation events (i.e., flights, truck deliveries).

These transportation events are characterized by a unique schedule Number, a type (e.g, flight, truck), and a delivery Route.

Please create an Entity Relationship diagram that captures this information about the UPS system.

Be certain to indicate identifiers and cardinality constraints.